FarmFederate — Supplementary Figures

Every figure here is generated from a completed run whose aggregate numbers are already reported in the main paper: the aligned 222/74/75 crop–note support, the clubbed centralized/federated sweep, the matched-setting standard tier (50 rounds), the three-seed frozen-encoder federated runs, and the advisory retrieval evaluation. Figures the main paper carries are not repeated. Plots from the superseded pipelines that remain in the repository — the `rag_01`–`rag_06` series and the earlier five-class crop-stress charts — measure a different corpus and are deliberately excluded.

I. Encoders on the Aligned Support

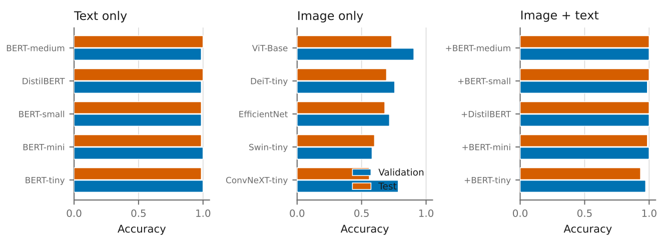

Fig. 1. Validation and test accuracy for every encoder on the identical 222/74/75 support. Text systems are saturated; the spread lives on the image side.

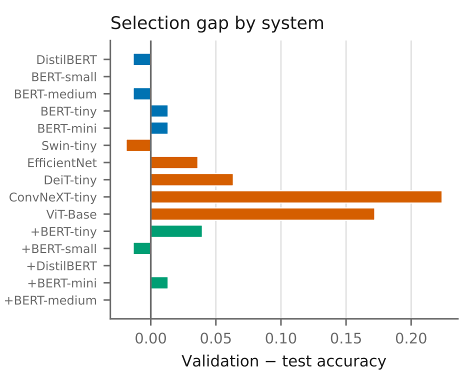

Fig. 2. Validation minus test accuracy per system. A large positive gap means the configuration was chosen on 74 crops and did not transfer; ConvNeXT-tiny and ViT-Base show the largest.

II. Note sparsity, per encoder

III. Federated behaviour

IV. Advisory corpus

V. Figures cut from the main paper for space

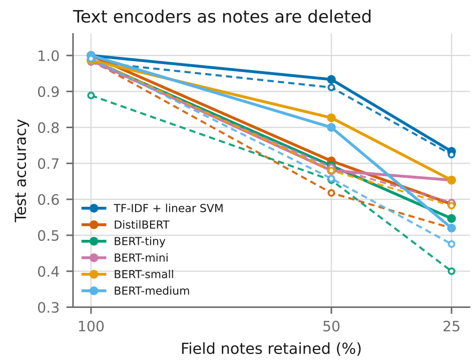

Fig. 3. Text encoders as notes are deleted; solid centralized, dashed federated. TF-IDF degrades most gracefully, which is the bag-of-words result discussed in the main paper.

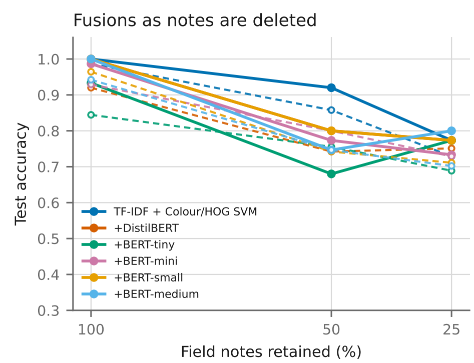

Fig. 4. Fusions as notes are deleted, same protocol. The fused systems hold above their text parents once a quarter of the notes remain.

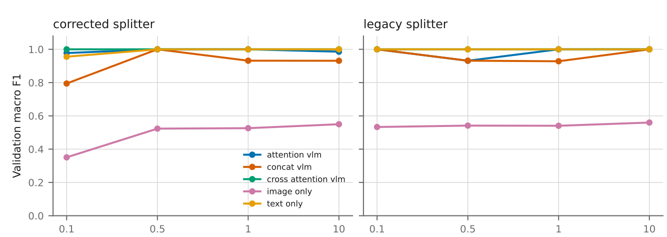

Fig. 5. All five architectures across the four skews under both splitters. The corrected class-wise splitter produces the intended label skew; the legacy size-only splitter does not, so its curves are flat in α.

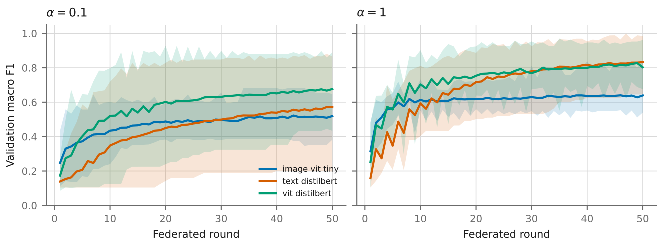

Fig. 6. Per-round convergence of the frozen-encoder systems at both skews, three seeds. The band spans the observed seed range.

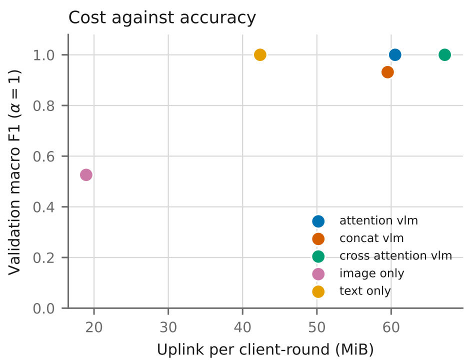

Fig. 7. Analytic float32 uplink per client-round against accuracy at $\alpha = 1$. The image-only arm is the cheapest and the weakest; text-only reaches the ceiling at 42.4 MiB.

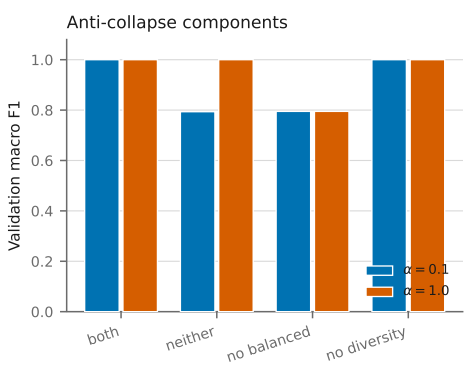

Fig. 8. Anti-collapse components under actual federation, concat model. No arm collapses, so the stack is reported rather than claimed necessary.

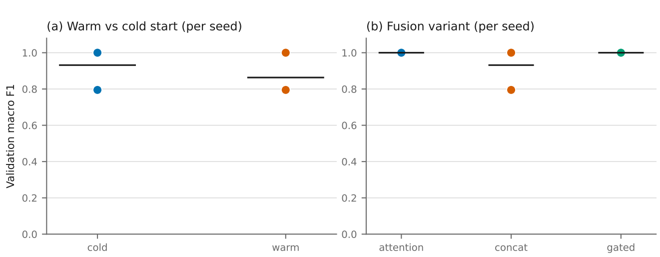

Fig. 9. (a) Warm versus cold initialization, per seed. Trained out to 50 rounds cold start is better at one seed and identical at the other, reversing the eight-round result. (b) Fusion variants across seeds.

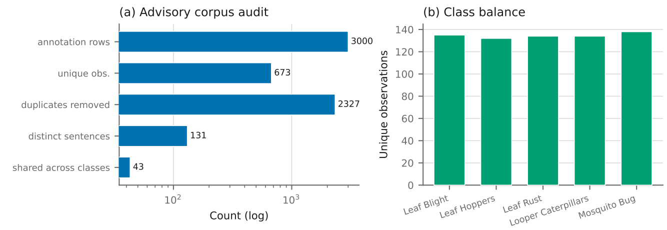

Fig. 10. (a) What the advisory corpus contains after de-duplication: 3000 annotation rows collapse to 673 unique observations over 131 distinct sentences, 43 of which appear in more than one class. (b) Class balance of the unique observations.

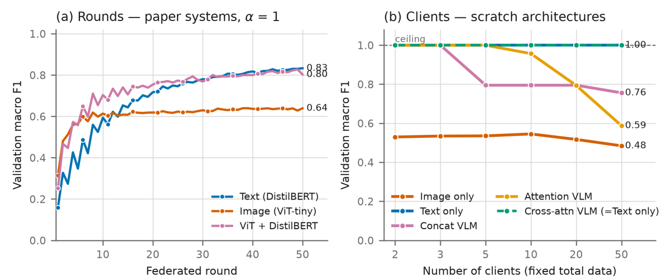

Fig. 11. Federated rounds and client scaling. Text overtakes fusion at round 33; text-only and cross-attention hold the ceiling to $K = 50$.

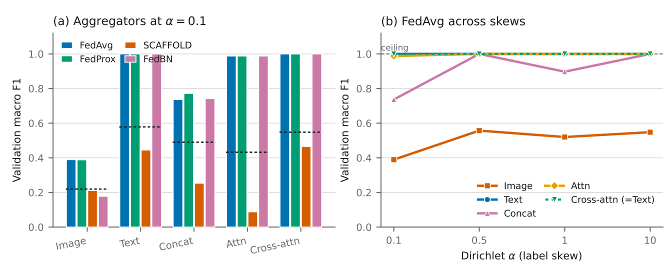

Fig. 12. Aggregators at $\alpha = 0.1$ with the local-only control, and FedAvg across skews. SCAFFOLD collapses to 0.088 on the attention VLM.

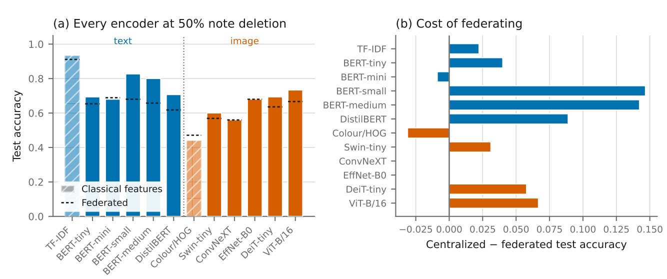

Fig. 13. Per-encoder detail at 50% note deletion, and the cost of federating each system.

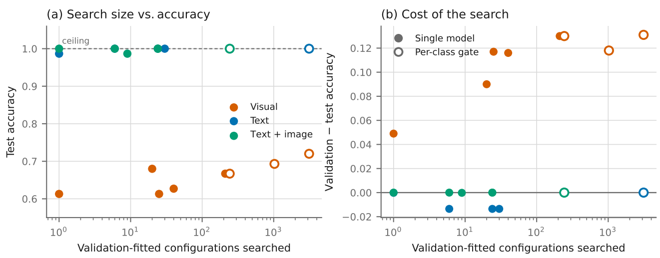

Fig. 14. Pipeline search size against accuracy, and the validation-to-test gap it buys. Gates are hollow.

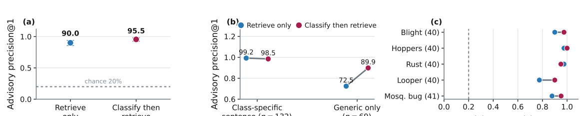

Fig. 15. Classify-Retrieve-Advise quality: headline precision@1, the class-specific versus generic split, and the per-class breakdown.